

Computer Science with Artificial Intelligence

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BIOMEDICAL LEARNING PLATFORM

by

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INTRODUCTION

Project Brief

The biomedical learning platform is a modernisation of a legacy Windows XP based educational system used for biomedical training and assessment. It's objective is to rebuild the platform as a secure, modern, web based application with improved UI.

This platform is intended to serve students and educators by offering interactive modules, quizzes, analytics and content management tools accessible across devices through the internet.

Requirements

1. Modernise the legacy Windows XP application into a web based platform.
2. Improve user experience with a clean UI, responsive design and accessibility compliance.
3. Allow interactive learning.
4. Create a scalable backend for user authentication, content management and analytics.
5. Provide instant feedback and support via an AI chatbot.
6. Support role based access control (Admin / Teacher / Student).
7. Ensure long term maintainability, modularity and extensibility.
8. Support media rich biomedical content (images, diagrams, 3D, videos).

Literature Reviews

A learning platform is not a new idea, but I believe all current solutions are lacking. After some market research on my competitors this is what I discovered;

Notion begins as an empty workspace where users can organise notes, databases and tasks with a high level of customisation. This appeals to a lot advanced users as it's extremely flexible. However, it does not offer any structure or concept of mastery for learners to achieve. This results in learners spending a lot of time organising content rather than actually learning and understanding content.

Anki is an effective spaced repetition software tool (flashcard) that has been developed to help long term memory through active recall. Whilst it may be very effective for memorisation; it lacks context, conceptual models and an encouraging user experience. The application has an immense learning curve and lacks guidance, causing many people to give up on it despite its effectiveness.

Udemy is a large marketplace for courses holding thousands of video based courses, most available at an affordable price. Although it excels in volume and accessibility, learning on Udemy is mainly passive. There's limited interaction, feedback, or personalisation, resulting in low completion rates and learners that often finish courses without achieving measurable mastery.

Khan Academy allows for concept driven learning with explanations and practice. Its biggest strength is in the introduction of concepts and its biggest weakness is its rigid learning pathways and inability to adapt to each student. This is because Khan Academy fails to allow continued learning after the introduction to the concept.

LinkedIn Learning is all about professional development and skills for the workplace, this is done using video learning courses. Though it benefits from great branding and an excellent interface, the learning experience is rather passive with mastery not being the focus. It's often used for awareness, not learning.

From the available learning solutions, a gap in what is delivered and what is effectively learned always seems to occur. Solutions either offer flexibility without any guide or structure without adaptation. This leads to passive engagement and low mastery. My platform bridges this gap by bringing together for structure, active learning and adaptive progression into a single system that is learner first, renews old learning models and allows for meaningful and measurable learning outcomes. Furthermore, AI would be used to help teachers generate quizzes and give immediate feedback to students.

Questionnaire

The questionnaire was used to assess the previous learning platform that was used on windows XP. Its purpose was to gain information on the experiences that the users had; the questions covered what the users liked about the learning platform, what problems they had and what could be improved. The information gathered will be very helpful and provided a basis for making recommendations regarding how to improve the learning platform.

How easy was it to use the platform? *				
1	2	3	4	5
☐	☐	☐	☐	☐

Was the level of difficulty reasonable? *				
1	2	3	4	5
☐	☐	☐	☐	☐

How visually appealing was the platform? *				
1	2	3	4	5
☐	☐	☐	☐	☐

Risk And Mitigation

There are many delicate variables involved in this project likely to impact delivery, scalability and user experience. Early consideration of these can help mitigate each risk by focusing on design decisions with phased implementation, ensuring a successful transition from a static, XP model to a modern, learner centred platform.

What feedback would you like after questions? *

Short answer text

Did you encounter any technical difficulties? *

Short answer text

Which features were the most helpful? *

Short answer text

Which features were the most confusing? *

Short answer text

Are there any features you want to see added?

Short answer text

Any other feedback?

Short answer text

Adding interactions and behaviours into the system makes both the system design and development quite complex. The chance of developing an over engineered system, which leads to delays in system delivery, is addressed by setting and enforcing a tight minimum viable product, keeping early adaptive logic to rule based systems and focusing on key learning functionality rather than extended features.

Learning paths can be interpreted as restrictive if they are compared against open ended tools or passive course sites. This can lead to disengagement if the learner feels they are being controlled as opposed to the experience being guiding or facilitating them. This can be addressed by allowing the user the option to disregard learning paths and explore by themselves.

Neither legacy content nor infrastructure was designed to support modular, interactive, or adaptive learning. Converting static material and integrating the back end, will slow down development and introduce technical debt in its own right. This risk is mitigated by standardising content into reusable learning blocks, prioritising high impact content for conversion, isolating legacy systems behind clear interfaces or even completely replacing it and migrating functionality incrementally rather than trying to replace it all in one go.

Adaptive learning involves monitoring learning progressions or decision logic, so scaling can be a problem. Learner state models that are not well managed may affect the responsiveness or reliability of the systems. This can be alleviated by keeping data simple and discrete concerning mastery states, reducing real time processing, or developing systems with horizontal scalability from the onset.

Evaluation Structure

The final evaluation will aim to conduct a comprehensive appraisal of the effectiveness of the newly developed learning platform based on the project's requirements. The intention is to establish whether the developed solution has managed to adequately respond to each problem areas the previous platform rose, whilst improving user experience and optimising learning outcomes. The final evaluation will also offer elements of evidence regarding both aspects and areas for improvement related to the platform.

This assessment will be organised on several key factors; including usability, engagement, the quality of content, interactivity and overall user satisfaction. This is in an effort to evaluate the extent to which the platform delivers on the expectations of the learning audience, ensuring that all aspects of the given platform are considered.

Multiple methods will be used to collect data on the final evaluation. Quantitative methods like ratings, results and completion times will enable the measurement of the level of the performance on the platform. Then a qualitative study can be done through survey or interview tools that will enable an in depth look at user experience, preference and recommendations on how the platform can be improved.

The results from the final evaluation will be analysed and organised to emphasise the strengths, weaknesses, so I can easily suggest improvements for the future. This systematic approach makes it possible to not only confirm improvements already made but also develop the system further to successfully serve users.

DESIGN

MoSCoW Analysis

Must modernises the Windows XP legacy app to a web platform, as it is necessary to make it accessible, secure and viable over the long term. A good user experience is ensured by having a clean UI, responsive design and accessibility compliance to facilitate learning effectively. This aligns with universal design for learning (UDL), a framework that optimises teaching and learning for all people based on scientific insights into how humans learn.

Interactive learning is necessary to shift the paradigm of learning from passive to active learning. The theory of learning is established on the theme of constructivist learning theory, which confirms that it is better for students to learn by actively building their knowledge instead of just receiving it. Specifically, this learning platform would benefit from the active recall method, based on the findings of the Testing Effect study, which states it is better to test a user to recall data from memory instead of reading it.

Furthermore, to provide support for user logins, content management, and data analytics needed to grow this platform, a scalable backend must be developed. To create distinct levels of access for administration, teaching, and students with restricted levels of access for all users to ensure that not all users possess the same levels of access to all functions, role based access control (RBAC) must be used. Following the principle of least privilege (PoLP), users must obtain restricted levels of access to all functions to complete specific tasks with limited capacity to affect this system.

Should provide media rich content like images, diagrams, video, and 3D graphics, since these greatly enhance the effectiveness of learning. By using visual and verbal channels side by side, the system takes the dual coding theory of Paivio (1971), which states that it creates multiple ways of retrieving information. On the other hand, the presence of multimedia strongly supports Mayer's multimedia principle (2014) stating, "People understand an explanation better if words are presented as speech combined with illustrations rather than as text combined with illustrations". This is even more crucial in biomedical education because most complex scientific concepts greatly exceed the limits of purely verbal instruction.

Focus should also be on ensuring that the platform can be maintained in a sustainable manner in the future years, this can be done by being modular and extensible. These requirements provide tremendous value to the platform but are not considered crucial from an initial implementation stage.

Could have instant feedback services and quiz generation through an artificial intelligence chatbot. Although adding this feature would be helpful in increasing the user's experience, the system can still operate effectively

without this feature in its early stages. This makes it a could requirement as it's a desirable feature that augment the functionality but can be postponed if there is a time or resource constraint

Human Computer Interaction

The key goal of an interface design is to ensure the cognitive load, amount of mental work required by the user to accomplish their task, is minimised. This can be done by mapping the site's navigation to already established mental models, guaranteeing that customers are able to apply their knowledge of other systems to our own and predict how it would work. Our intuitive interface accomplishes this by making strategic placements of interface elements such as the navigation menu and the search box, so customers can easily concentrate on the data without having to concentrate on how to operate the interface.

The design is consistent with Fitts' Law, where the time necessary for making a move towards a target is a function of the distance of the target and its size. Within the context of this website, this has meant the strategic use of call to action buttons as well as optimizing click zones, with high priority functions having larger surface areas situated in accessible areas of the viewport. This is especially true if the visitor is using a mobile touch interface.

An essential part of the user experience is the use of immediate feedback in the system. TheHeuristics by Jakob Nielsen states that the system must always inform users of what is happening through proper feedback in a reasonable time frame. The design principles use hover states, loaders, and success animations to maintain a dialogue between man and machine. It is essential to establish user control through these micro interactions, as they validate the user's activities.

Login Page

The login page should allow users the option to sign in with their username to track their progress, or anonymously as a guest.

Welcome to the
biomedical learning platform!

Username

password

Home View

The home view should allow admins and teachers to navigate to their respective pages. Also, it should welcome users to the site as after logging in this is the page that everyone is redirected to. It holds all the quizzes and can be filtered by class, topic and title.

admin teacher username logout

Welcome to the homepage

search for quiz

quiz 1 heart features (10q)	quiz 2 leg muscles (6q)
quiz 3 general knowledge (20q)	quiz 4 Biomechanics (10q)

Quiz View

When a quiz is selected, they are taken to here. This page is where the user does their learning and there's a chatbot at the bottom for if they need help mid question.

The sketch shows a user interface for a quiz. At the top, there are navigation links: 'home', 'admin', 'teacher', and 'logout'. Below these is the title 'Quiz 1'. A box contains the question 'question 4/10' and 'Which is not a part of the heart?'. Below the question are four multiple-choice options: (a) Vena Cava, (b) Aorta, (c) Medulla, and (d) Pulmonary Vein. At the bottom of the box are two buttons: 'Previous' and 'Next'. Below the box, there is a 'Need help' label followed by a text input field.

Teacher View

Teacher view is accessible to both admins and teachers, everyone should be able to navigate to home page but only admins to the admin view. It contains quiz analytics, such as average score, average time, total attempts and hardest question. It also allows teachers to view previous quizzes and create new ones, furthermore they can manage their classes by adding new quiz to a specific class or removing redundant ones.

Also, there is a field where teachers can enter a topic and AI will generate a quiz from the existing question pool.

home admin username logout

Analytics

All

Student

quiz

avg score avg time totl attempts hardest q
72% 15s 20 ID:145

Create new quiz

Topic

title

description

add question

generate quiz

View quizzes

quiz1

view

edit

delete

quiz2

view

edit

delete

Manage class

quiz1

add to class

remove from class

Create question

Title

Type

Options

Answer

Submit

Admin View

Admin view is the most restricted page, only admin roles can access this. It should be able to navigate to both teacher and home page. It holds a table where you can reassign other users' roles.

home teacher

username logout

enter search term

Username	email	current role	change role
User 1	email@1	teacher	teacher
User 2	email@2	student	student
User 3	email@3	admin	admin

Databases

Not including Supabase's authentication tables, have 4 custom ones each normalised to provide quiz and non redundant navigation.

<u>Quiz</u>					
Quiz ID	Topic	Title	Description	Creator	Question List

<u>Class</u>			
Class ID	UserList	Teacher ID	Quiz ID

<u>Quiz Questions</u>						
Question ID	Quiz ID	type	text	options	answer order	image

<u>Quiz Analytics</u>					
Analytics ID	Question ID	User ID	correct	timeSpent	date

IMPLEMENTATION

Frontend

The frontend was implemented using a technology stack of React, TypeScript and JavaScript. This allows for a modular approach, enabling the learning content to be broken down into interactive lessons, quizzes and dashboards.

The frontend handles user interaction with the custom backend API routes and Supabase services. The responsiveness of the platform ensures compatibility with desktop, tablet and mobile platforms and this is a major improvement over the static GUI of the legacy system.

Backend

The custom backend functions were created using Node.js. These include RESTful api routes that are responsible for any application specific functionality not handled by supabase; such as analytics, complex interactions.

The backend routes interact with Supabase to control data access and user authentication, ensuring a safe and dynamic link between the frontend and backend as service solution.

Database

Supabase was employed as the backend as service solution for the platform, which takes care of secure storage, authorisation and API tasks. The communication with Supabase is made by both the frontend and backend, they fetch and update the learning material whilst also tracking user data. The idea is that no custom database queries were written in the application so Supabase would handle tasks on its own, allowing for scalable and reliable service.

Interactive Features

Interactive features include dynamic lessons, quizzes and progress monitors that adjust in real time based on user input. Students are provided instant feedback to promote engagement. Performance metrics such as average scores, time taken in quizzes and error mistakes are also tracked to monitor student performance, enabling both students and educators to monitor progress and areas for improvement based on subject matter retention.

AI Chatbot

A Google Gemini powered AI chat bot was implemented for immediate support for learners. The AI powered bot is used for content queries, navigation and overall assistance, reducing the need for instructor support.

Development Tools and Version Control

Development work was performed in a Visual Studio Code environment, with GitHub providing the version control functionality for the management of the source code. The environment for building both the frontend and running the backend APIs had been Node.js.

Challenges

In development, various problems were encountered with the implementation of interactive learning features. First, I had investigated the use of H5P to provide rich interactive content. However, the free version of H5P was not powerful enough for supporting more complex interactive features like clickable images or hotspot style questions. I also tried to create and edit quizzes in a reliable, flexible way by allowing learners or instructors directly within the application via iFrame based approaches, this also did not work.

Due to this, I have implemented a completely custom interface for both creating and rendering quizzes, giving me full control over interactive elements and question types. This approach allowed for all features to be natively integrated into the platform, maintaining responsiveness and intuitiveness for users.

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